CSE556 Natural Language Processing

Assignment 3: Build a Mini Hindi-GPT From Scratch

Group no: 28

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1. Work Distribution

With major contributions of Vaibhav in Task 1, Reehan in Task 2 and Vishal in Task 3. The entire pipeline from data preprocessing through model architecture, training, evaluation, and report writing was improved upon jointly. All members are familiar with all parts of the assignment.

2. Plots and Metrics

Task 2

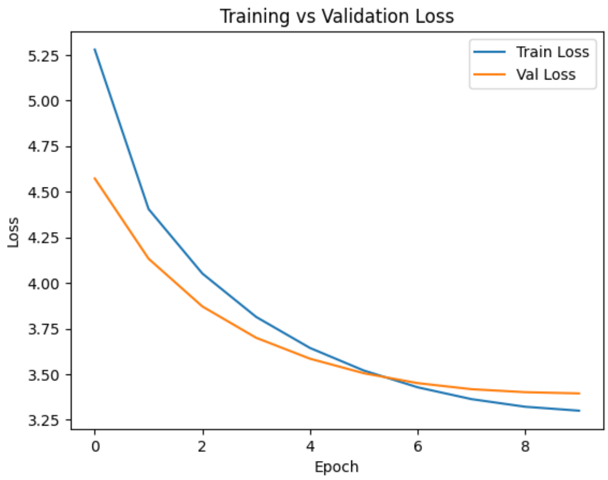
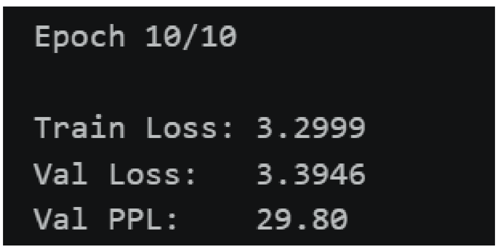


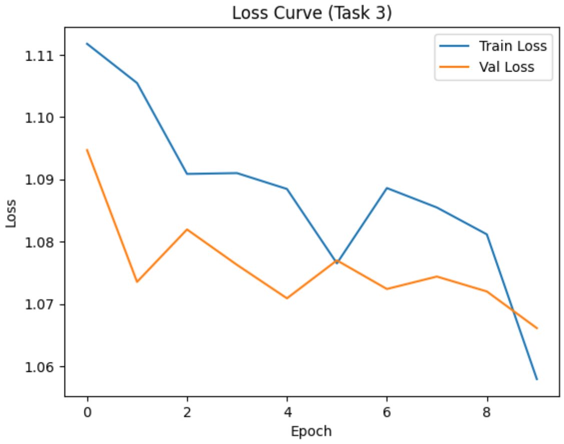
Figure 1: Language Model Loss Curve

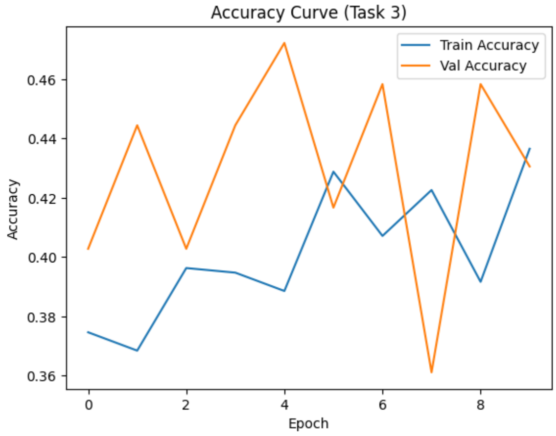
The language model was trained for 10 epochs with early stopping and checkpointing. Both training and validation loss decreased consistently, indicating effective learning. The validation perplexity improved steadily from **96.81 to 29.80**, demonstrating significant gains in language modelling capability.

- **Best Model Val Loss**: 3.3946
- **Best Model Val Perplexity:** 29.80



Task 3





The classifier was trained for up to 15 epochs using the pretrained GPT backbone with early stopping. The model demonstrated stable training behavior, with gradual improvement in both training and validation metrics.

The best performance was achieved in the 5th epoch with:

- **Validation Loss:** 1.0661
- **Validation Accuracy:** 47.22%

Compared to alternative configurations, this model showed more consistent validation performance and lower validation loss, indicating better generalization and stability. Here we considered the last 512 tokens of the text as they seem to carry the most meaning of the review like text.

```
Epoch 5/15
Training: 100%|██████████| 21/21 [00:05<00:00,  3.74it/s, loss=1.1000, acc=0.3885]
Evaluating: 100%|██████████| 3/3 [00:00<00:00,  7.35it/s]
Train Loss: 1.0884
Train Acc:  0.3885
Val Loss:   1.0709
Val Acc:    0.4722
```

Although a peak validation accuracy of **50.0%** was observed using a different strategy where we used first 512 tokens of the text but that model exhibited unstable training behavior and higher validation loss. Therefore, the more stable model was selected for final evaluation.

To achieve this, we froze all the weights of the pre-trained GPT Language Model except the last 2 layers (transformer blocks). We tried many variations of unfreezing layers, sometimes decreased or increased the learning rate, tried 2 different block sizes of 256 and 512 with the latter proving to be yield better performance in classifier model.

3. Additional Information

AI tools have been used to do research on improvement of the model along with generation of the code parts.